

Swetha Siripurapu

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SUMMARY

Computer Science undergraduate with hands-on experience building AI-powered and cloud-based applications using Python, Java, AWS, Docker, and SQL. Experienced in Computer Vision, Generative AI, LLMs, and REST APIs, with a strong focus on developing practical, real-time intelligent solutions. Strong foundation in Data Structures / Algorithms, Object-Oriented Programming, DBMS, Operating Systems, and Computer Networks, with 700+ coding problems solved.

PROJECTS

AI-Based Helmet Detection System

[GitHub](#)

- Built an AI-powered real-time helmet detection system using Python, YOLOv8, OpenCV, and Roboflow for live video inference and violation detection.
- Built an end-to-end Deep Learning inference pipeline for detecting helmet violations from live video streams.
- Implemented real-time Text-to-Speech (TTS) voice alerts using pyttsx3 for helmet violation detection
- Integrated Amazon S3 using boto3 to automatically store captured violation images.
- Implemented SMTP-based email notifications with Amazon S3 image URLs for administrators.
- Built a Streamlit dashboard to visualize detection logs, timestamps, and captured images.

ICU Face Recognition Access System

[GitHub](#)

- Developed an AI-powered face recognition system for secure visitor authentication using AWS Rekognition, S3, and backend APIs.
- Captured and processed visitor images for real-time identity verification.
- Integrated AWS S3 and Rekognition for secure image storage and face matching.
- Designed backend APIs and business logic for visitor validation, authentication, and access control.
- Generated Amazon SNS notifications for unauthorized access attempts.

EXPERIENCE

AWS & DevOps Intern, Technical Hub Pvt. Ltd.

May 2025 - Jun 2025

- Worked with AWS services including EC2, S3, IAM, Lambda, and CloudWatch for deploying cloud applications.
- Automated deployment workflows using Git, Docker, and CI/CD concepts.
- Collaborated in an Agile development environment following SDLC practices.
- Assisted in cloud infrastructure deployment, monitoring, and system automation.
- Worked closely with developers to improve deployment efficiency and application reliability.

TECHNICAL SKILLS

Programming Languages:	Python, Java, C
Generative AI & Machine Learning:	Generative AI, LLM, Prompt Engineering, Computer Vision, Deep Learning, YOLOv8, OpenCV, AWS Rekognition, AWS Comprehend, AWS Textract
Backend & APIs:	REST APIs, HTTP/HTTPS, API Integration
Cloud & AWS:	EC2, S3, IAM, Lambda, VPC, API Gateway, RDS, Elastic Beanstalk, Load Balancer, CloudWatch
DevOps:	Git, GitHub, Docker, Jenkins, Kubernetes (Basics), CI/CD
Databases:	SQL, MySQL
Web Technologies:	HTML, CSS, JavaScript
Core Subjects:	Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks
Operating Systems:	Linux, Windows

CERTIFICATIONS

- **AWS Certified Developer - Associate**
- **Red Hat Certified System Administrator (RHCSA)**
- **Oracle Certified Foundations Associate** – Java Certified Foundations Associate
- **Oracle Certified Foundations Associate** – Database
- **IT Specialist – HTML & CSS** (Certiport, Pearson, CertNexus)

ACHIEVEMENTS

- Solved **700+ coding problems** across LeetCode, CodeChef, and HackerRank.
- Earned a **5-Star Gold Badge** in SQL and **3-Star Silver Badges** in C, Java, and Python on HackerRank.
- Selected among the **Top 50 out of 600+ participants** in Technical Hub's SQL Query Competition and awarded an **Oracle Database certification voucher**.

EDUCATION

BTech in Computer Science , Aditya College of Engineering and Technology	Expected 2027 CGPA: 8.92
Intermediate , Sri Chaitanya Junior College MPC	2021 – 2023 CGPA: 97.1